

# Eprad eCNA Cinema Automation Plugin

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Version 1.1 | Build 34

Use this plugin to monitor and control an Eprad eCNA Cinema Automation system from within your Q-SYS designs. The plugin communicates over two independent channels: the CAI (Cinema Automation Interface) ASCII TCP protocol for commands and status, and a direct HTTP connection for I/O flag detail.

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## Compatibility

This plugin was developed and tested with Q-SYS Designer Software version 9.x and Eprad eCNA firmware v1.23 and v1.26. It is designed to work with all eCNA models.

**Note:** The plugin communicates via the CAI TCP interface (default port 13002) and the eCNA HTTP web interface (port 80). Both must be reachable from the Q-SYS Core on your network. Because the eCNA supports only one CAI TCP connection per port at a time, avoid using the same port number in multiple concurrent designs.

## Configuration Overview

### Connect the Plugin

1. Drag the Eprad eCNA plugin into your Q-SYS schematic.
2. In the plugin Properties, configure the number of Macros and RDI devices for your installation. Enable Show Debug Tab only during commissioning.
3. Save the design to your Core and run (F5).
4. On the Setup page, enter the IP address of the eCNA and confirm the TCP Port (default 13002).

The plugin connects automatically once a valid IP address is entered. A Connection Status of **OK** confirms a successful connection.

### Communication Architecture

The plugin uses two simultaneous communication channels:

- **CAI TCP (port 13002):** The primary control channel. Sends commands (EVT, OUT, RDI, MAC) and receives status responses (RST, RID) and unsolicited event reports (RPT,EVT). Polled at the RST Poll Rate interval as a safety net; most status updates arrive as unsolicited events. Only one device may have a port open at a time.
- **HTTP (port 80):** A secondary read-only channel for I/O flag detail, providing the state of all 44 output flags and 31 input flags, including house lights, curtain, masking, lens, sound, and aux outputs. Polled independently at the Web Scrape Rate.

**Note:** The HTTP scrape does not consume a CAI port. Multiple Q-SYS designs can scrape the eCNA web interface simultaneously without conflict. Only the CAI TCP connection is exclusive per port.

## Unsolicited Event Reports

The plugin subscribes to unsolicited event reports from the eCNA (CFG, RPTON) on connection. The eCNA sends RPT, EVT messages when key state changes occur (START, STOP, CUE, projector power changes, fire stop, etc.). The plugin triggers an immediate RST status re-poll on receipt of these events, providing near-instant UI updates without relying solely on the poll timer.

## Properties

### Num Macros

Integer, 0–50 (default 50). Sets the number of Macro trigger buttons available on the Macros tab. Set to 0 to hide the Macros tab entirely.

### Num RDI

Integer, 0–5 (default 5). Sets the number of RDI (Remote Device Interface) devices. Each device has 16 command buttons. Set to 0 to hide the RDI tab entirely.

### Poll Rate

Integer, 1–60 seconds (default 30). Interval in seconds between RST status poll commands sent over the CAI TCP connection. Because the plugin subscribes to unsolicited event reports, most real-time updates arrive without waiting for this timer. This rate primarily serves as a safety net to correct any state drift. Values of 30–60 seconds are appropriate for production use.

### Scrape Rate

Integer, 1–60 seconds (default 3). Interval in seconds between HTTP requests to the eCNA web interface for I/O flag detail. This is the primary update mechanism for the output and input flag indicators (house lights, curtain, masking, lens, sound, aux outputs, and input flags). Values below 3 seconds may interfere with direct operation of the eCNA device. The original script used 1.3 seconds; 3 seconds is recommended.

### Auto Refresh Names

Boolean (default on). When enabled, the plugin periodically re-fetches input and output flag names from the eCNA web interface, alternating between the two pages roughly every 30 seconds so each page

refreshes approximately once per minute. Disable this if you do not want background name refreshes; names will still be fetched once on initial connection. Names can always be refreshed manually using the buttons on the Inputs and Outputs pages.

### Show Debug Tab

Boolean (default off). When enabled, adds a Debug tab to the plugin showing Last TX, Last RX, and Last Error fields. Also enables verbose output to the Q-SYS debug window (visible via Help > Q-SYS Debug) showing all CAI commands sent, responses received, and status changes. Disable for production deployments.

## Controls

### Setup Page

#### IP Address

Text entry. Enter the IP address of the eCNA. The plugin connects automatically when a valid address is entered or changed.

#### TCP Port

Text entry. The CAI TCP port number (default 13002). Change this to match the CAI channel assigned to this Q-SYS design. Common values: 13000 (channel 1), 13001 (channel 2), 13002 (channel 3), 13003 (channel 4).

**Note:** Port 13003 (channel 4) requires eCNA firmware version 1.25 or later.

#### Status LED / Status

Indicator LED and text field, read only. Displays the current connection state:

- **OK** (green): Connected and communicating.
- **Initializing** (blue): Connecting or establishing initial communication.
- **Disconnected / Fault** (red): Connection lost or error received.
- **Missing** (orange): No IP address configured.

#### Reconnect

Trigger button. Forces an immediate reconnection attempt. Useful if the connection has stalled, the IP address was changed, or the port was changed. Note: changing the TCP Port does not take effect until Reconnect is pressed.

#### Model

Text field, read only. Displays the eCNA device model returned by the RID command on connection.

#### Firmware

Text field, read only. Displays the eCNA firmware version returned by the RID command.

### Outputs Page

The Outputs page contains all eCNA output controls, grouped by type. Toggle buttons both send commands to the eCNA and reflect the current output state as returned by the web scrape.

### **Digital Projectors (Dig Proj1 Power, Dig Proj1 Video, Dig Proj2 Power, Dig Proj2 Video, Slide Projector)**

Toggle buttons. Control and indicate the power and video output state of the two digital projectors and the slide/auxiliary projector. State is updated from the web scrape (flags 1–4, 19).

### **House Lights**

Mutually exclusive Toggle buttons: None / Down / Mid1 / Mid2 / Up. Sends OUT HLxxx commands. State updated from scrape flags 5–8.

### **Stage Lights**

Mutually exclusive Toggle buttons: None / Down / Up. Sends OUT SLxxx commands. State updated from scrape flags 9–10.

### **Curtain**

Mutually exclusive Toggle buttons: None / Close / Open. Sends OUT CURTxxx commands. State updated from scrape flags 11–12.

### **Masking**

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends OUT MSKxxx commands. State updated from scrape flags 13–15.

### **Lens**

Mutually exclusive Toggle buttons: None / Flat / Scope / Special. Sends OUT LENxxx commands. State updated from scrape flags 16–18.

### **Sound**

Mutually exclusive Toggle buttons: None / Aux1 / Aux2 / Dig1 / Dig2 / Mono / NonSync / SR / SVA. Sends OUT SNDxxx commands. State updated from scrape flags 20–28.

### **Mute**

Independent Toggle button. Sends OUT MUTEON or OUT MUTEOFF. State updated from scrape flag 25.

### **Aux Output Flags 1–16**

Toggle buttons arranged in two columns of eight. Each row shows the flag number (Out 1–Out 16), the toggle button, and the user-defined name fetched from the eCNA. Send OUT OUTnON/OFF commands. State updated from scrape flags 29–44.

**Note:** Names refer to Output Flag names as configured in the eCNA web interface, not physical output names on any I/O board.

### **Refresh Output Names**

Trigger button. Immediately refreshes the current output flag names from the eCNA. The names will update within a few seconds. Available regardless of the Auto Refresh Names property setting.

## Inputs Page

Displays all 31 eCNA input flags as LED indicators. Two columns: General Inputs (In1–In16) on the left and System Inputs on the right. Each General Input row shows the LED, the flag number (In1–In16), and the user-defined name fetched from the eCNA.

**Note:** Names refer to Input Flag names as configured in the eCNA web interface, not physical input names on any I/O board. Input flags are read-only indicators; they cannot be set from the plugin.

### Refresh Input Names

Trigger button. Immediately refreshes the current input flag names from the eCNA. The names will update within a few seconds. Available regardless of the Auto Refresh Names property setting.

## Macros Page

Visible only when Num Macros is greater than 0. Contains trigger buttons for Macros 1 through N, arranged 10 per row. Each press sends `EVT MAC, n` to the eCNA. Text labels appear above each button.

## RDI Page

Visible only when Num RDI is greater than 0. Contains trigger buttons for up to 5 RDI (Remote Device Interface) devices, each with 16 commands (labelled RDI 1-1 through RDI 5-16), arranged in two rows of 8 per device. Each press sends `EVT RDI, device, command` to the eCNA. Text labels appear above each button.

## Program Page

### Transport Controls (Start, Stop, Cue, Abort)

Trigger buttons. Send `EVT STA`, `EVT STP`, `EVT CUE`, and `EVT ABT` commands respectively.

### Fault Controls (Set Fault, Clear Fault, Reset Fault, Cancel Alarm)

Trigger buttons. Send `EVT FLT`, `EVT CLR`, `EVT RES`, and `EVT CNL` commands respectively.

### Program # / Set Prog / Set & Run

Combo box and trigger buttons. Select a program number (1–9) and press Set Prog to send `EVT PGM, n`, or Set & Run to send `EVT SPR, n` (set and immediately start).

### Control State

Text field, read only. Displays the current control state from the RST response: **IDL** (idle) or **RUN** (running).

### Stopped State

Text field, read only. Displays the stopped state: **OK**, **STP** (stopped), **FLT** (fault), or **FIR** (fire stop active).

### Cue Number

Text field, read only. Displays the current cue number within the running program.

### Program Number

Text field, read only. Displays the currently active program number.

### Segment Name

Text field, read only. Displays the name of the current program segment.

### Segment Index

Text field, read only. Displays the current segment index within the program.

### CMD Access

Text field, read only. Displays the CAI command access status: **OK** (full access) or **BLK** (blocked — the eCNA is not accepting CAI commands).

## Debug Page

Visible only when Show Debug Tab is enabled in Properties.

### Last TX

Text field, read only. Displays the last CAI command string sent to the eCNA (without the trailing CR/LF).

### Last RX

Text field, read only. Displays the last response line received from the eCNA.

### Last Error

Text field, read only. Displays the most recent error message, including CAI errors (RSP,ERR), HTTP scrape errors, and internal runtime errors.

## Input & Output Flag Names

Version 1.1 adds the ability to display user-defined names for the 16 General Input flags (In1–In16) and the 16 Aux Output flags (Out1–Out16). These names are configured on the eCNA web interface under Setup → Input Flag Names and Setup → Output Flag Names.

### How Names Are Fetched

Input and output flag names are retrieved automatically from the eCNA on initial connection. When Auto Refresh Names is enabled, names are refreshed approximately once per minute in the background with minimal impact on normal operation.

### Manual Refresh

The Refresh Input Names button on the Inputs page and the Refresh Output Names button on the Outputs page each immediately refresh the relevant names from the eCNA, typically appearing within a few seconds. These buttons work regardless of the Auto Refresh Names setting.

### Where Names Appear

- **Inputs page:** Each General Input row (In1–In16) shows the LED indicator, the flag number label, and the user-defined name in a text field alongside it.

- **Outputs page:** Each Aux Output Flag row shows the flag number, the toggle button, and the user-defined name in a text field to the right of the button.
- **Control pins:** In Names 1–16 and Out Names 1–16 expose the names as readable output pins for use elsewhere in the Q-SYS design.

**Note:** Names reflect the flag names as configured in the eCNA, not the physical label on any I/O board terminal. The eCNA supports up to 14 characters per name.

## Control Pins

The following table lists the control pins available when this plugin is used in a Q-SYS schematic. All pins are optional; connect only those required by your design.

| Pin Name                  | Value                      | String                                                                                                             | Position | Direction |
|---------------------------|----------------------------|--------------------------------------------------------------------------------------------------------------------|----------|-----------|
| <b>Setup</b>              |                            |                                                                                                                    |          |           |
| IP Address                |                            | (text)                                                                                                             |          | Input     |
| TCP Port                  |                            | (text)                                                                                                             |          | Input     |
| Connection Status         | 0<br>1<br>2<br>3<br>4<br>5 | OK (Green)<br>Compromised (Yellow)<br>Fault (Red)<br>Not Present (Grey)<br>Missing (Orange)<br>Initializing (Blue) | —        | Output    |
| Status LED                | 0–5                        | (same as Status)                                                                                                   | —        | Output    |
| Connected                 | 0 / 1                      | Off / On                                                                                                           | 0 / 1    | Output    |
| Reconnect                 | trigger                    | —                                                                                                                  | —        | Input     |
| Model                     |                            | (text)                                                                                                             |          | Output    |
| Firmware Version          |                            | (text)                                                                                                             |          | Output    |
| <b>Program State</b>      |                            |                                                                                                                    |          |           |
| Control State             |                            | IDL / RUN                                                                                                          |          | Output    |
| Stopped State             |                            | OK / STP / FLT / FIR                                                                                               |          | Output    |
| Cue Number                |                            | (text)                                                                                                             |          | Output    |
| Program Number            |                            | (text)                                                                                                             |          | Output    |
| Segment Name              |                            | (text)                                                                                                             |          | Output    |
| Segment Index             |                            | (text)                                                                                                             |          | Output    |
| CAI CMD Access            |                            | OK / BLK                                                                                                           |          | Output    |
| <b>Transport Commands</b> |                            |                                                                                                                    |          |           |
| Cmd Start                 | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Stop                  | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Cue                   | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Abort                 | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Fault                 | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Clear Fault           | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Reset                 | trigger                    | —                                                                                                                  | —        | Input     |
| Cmd Cancel Alarm          | trigger                    | —                                                                                                                  | —        | Input     |

| Pin Name                                                                      | Value                                     | String   | Position | Direction      |
|-------------------------------------------------------------------------------|-------------------------------------------|----------|----------|----------------|
| Set Program                                                                   | (text / combo box, 1–9)                   |          |          | Input          |
| Cmd Set Program                                                               | trigger                                   | —        | —        | Input          |
| Cmd Set And Run                                                               | trigger                                   | —        | —        | Input          |
| <b>Macros (1 – Num Macros)</b>                                                |                                           |          |          |                |
| Macros 1 ... Macros N                                                         | trigger                                   | —        | —        | Input          |
| <b>RDI (1 – Num RDI × 16)</b>                                                 |                                           |          |          |                |
| RDI 1 ... RDI 80                                                              | trigger                                   | —        | —        | Input          |
| <b>Outputs</b>                                                                |                                           |          |          |                |
| House Lights 1–5<br>(1=None 2=Down 3=Mid1 4=Mid2 5=Up)                        | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Stage Lights 1–3<br>(1=None 2=Down 3=Up)                                      | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Curtain 1–3<br>(1=None 2=Close 3=Open)                                        | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Masking 1–4<br>(1=None 2=Flat 3=Scope 4=Special)                              | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Lens 1–4<br>(1=None 2=Flat 3=Scope 4=Special)                                 | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Sound 1–9<br>(1=None 2=Aux1 3=Aux2 4=Dig1 5=Dig2 6=Mono 7=NonSync 8=SR 9=SVA) | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Mute                                                                          | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Dig1 Power                                                                    | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Dig1 Video                                                                    | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Dig2 Power                                                                    | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Dig2 Video                                                                    | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Aux Projector (Slide Projector)                                               | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| Aux Out 1–16                                                                  | 0 / 1                                     | Off / On | 0 / 1    | Input / Output |
| <b>Input &amp; Output Flag Names</b>                                          |                                           |          |          |                |
| In Names 1–16                                                                 | (text — user-defined name for In1–In16)   |          |          | Output         |
| Out Names 1–16                                                                | (text — user-defined name for Out1–Out16) |          |          | Output         |
| Refresh Input Names                                                           | trigger                                   | —        | —        | Input          |
| Refresh Output Names                                                          | trigger                                   | —        | —        | Input          |
| <b>Debug</b>                                                                  |                                           |          |          |                |
| Last Tx                                                                       | (text)                                    |          |          | Output         |
| Last Rx                                                                       | (text)                                    |          |          | Output         |
| Last Error                                                                    | (text)                                    |          |          | Output         |

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